

Precipitation

Answers

Observing precipitation

1. The following photograph shows the result of adding a colourless solution of lead nitrate to a colourless solution of potassium iodide in a test tube.



Photo: Lindsey Moore
Shutterstock.com

- (a) A yellow precipitate forms. Define the term 'precipitate'.
An insoluble solid forms when two solutions are mixed.
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- (b) The test tube is allowed to stand in a rack for 1 hour. What will happen to the precipitate in that time?
The precipitate will settle to form sediment at the bottom of the test tube.
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- (c) Name the compound making up the precipitate.
Lead iodide.....
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- (d) Write a word equation for the reaction.
Lead nitrate + potassium iodide → lead iodide + potassium nitrate.....

Predicting precipitation

2. The following table provides information on the solubility of various salts in water.

Anion	Soluble	Slightly soluble	Insoluble
Nitrate	All	None	None
Chloride	Most	PbCl ₂	AgCl
Sulfate	Most	CaSO ₄ , Ag ₂ SO ₄	BaSO ₄ , PbSO ₄
Carbonate	Na ₂ CO ₃ , K ₂ CO ₃ , (NH ₄) ₂ CO ₃	None	Most
Hydroxide	NaOH, KOH, NH ₄ OH, Ba(OH) ₂	Ca(OH) ₂	Most

Use this information to predict whether a precipitate will form when the nominated solutions are mixed. If a precipitate forms, name the precipitate.

- (a) Lead nitrate solution + sodium sulfate solution.
Lead sulfate will precipitate (as it is insoluble in water).
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- (b) Copper(II) sulfate solution + potassium hydroxide solution.
Copper (II) hydroxide will precipitate (as it is insoluble in water).
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- (c) Barium chloride solution + calcium nitrate solution.
No precipitate forms (barium nitrate and calcium chloride are both soluble in water).
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